

Lecture 03: Multivariate Probability

Introduction to Machine Learning

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Except explicitly cited, the reference for the material in slides is:

- Murphy, K. P. (2022). *Probabilistic machine learning: an introduction*. MIT press.

Section 1

Important Notation Definition

Notation for Random Variable, Vector and Matrix

Throughout the course, we use the following notation to show random variable, random vector, random matrix and their corresponding outcomes:

X	Random variable (Upper-case letter)
x	Outcome of a random variable (lower-case letter)
$\mathbb{X}$	Random vector/matrix (Blackboard boldface letter)
$\mathbf{x}/\mathbf{X}$	Outcome of a random vector/matrix (Boldface letter)
Θ	Random variable/vector/matrix
θ	Outcome of random variable
$\boldsymbol{\theta}$	Outcome of random vector/matrix

Section 2

Basic Definitions

Covariance

- Suppose two random variables X and Y . The Covariance is defined as:

$$\text{Cov}[X, Y] \triangleq \text{E}[(X - \text{E}[X])(Y - \text{E}[Y])] = \text{E}[XY] - \text{E}[X] \text{E}[Y]$$

- Assume $\mathbb{X} = [X_1, X_2, \dots, X_D]^T$ is a D-dimensional random vector, then its covariance matrix is defined as:

$$\begin{aligned} \text{Cov}[\mathbb{X}] &\triangleq \text{E}[(\mathbb{X} - \text{E}[\mathbb{X}])(\mathbb{X} - \text{E}[\mathbb{X}])^T] = \mathbf{\Sigma} \\ &= \begin{bmatrix} \text{Cov}[X_1, X_1] & \text{Cov}[X_1, X_2] & \cdots & \text{Cov}[X_1, X_D] \\ \text{Cov}[X_2, X_1] & \text{Cov}[X_2, X_2] & \cdots & \text{Cov}[X_2, X_D] \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}[X_D, X_1] & \text{Cov}[X_D, X_2] & \cdots & \text{Cov}[X_D, X_D] \end{bmatrix} \end{aligned}$$

- Cross-covariance: $\text{Cov}[\mathbb{X}, \mathbb{Y}] = \text{E}[(\mathbb{X} - \text{E}[\mathbb{X}])(\mathbb{Y} - \text{E}[\mathbb{Y}])^T]$

Covariance

- $\text{E}[\mathbb{X}\mathbb{X}^T] = \mathbf{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^T$, $\boldsymbol{\mu} \triangleq \text{E}[\mathbb{X}]$
- $\text{Cov}[\mathbf{A}\mathbb{X} + \mathbf{b}] = \mathbf{A} \text{Cov}[\mathbb{X}] \mathbf{A}^T$

Correlation

- Suppose two random variables X and Y . The Correlation that measure the level of **Linear** relation between two variables is defined as:

$$\rho \triangleq \text{Cor}[X, Y] \triangleq \frac{\text{Cov}[X, Y]}{\sqrt{V[X] V[Y]}}$$

- If $\mathbb{X}$ is a D -dimensional random vector, its correlation matrix is defined as:

$$\text{Cor}[\mathbb{X}] \triangleq \begin{bmatrix} \text{Cor}[X_1, X_1] = 1 & \text{Cor}[X_1, X_2] & \cdots & \text{Cor}[X_1, X_D] \\ \text{Cor}[X_2, X_1] & \text{Cor}[X_2, X_2] = 1 & \cdots & \text{Cor}[X_2, X_D] \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cor}[X_D, X_1] & \text{Cor}[X_D, X_2] & \cdots & \text{Cor}[X_D, X_D] = 1 \end{bmatrix}$$

Correlation

- One can show that $-1 \leq \rho \leq 1$
- $|\text{Cor}[X, Y]| = 1$ iff $Y = aX + b$

Correlation and Nonlinear Dependencies [1]

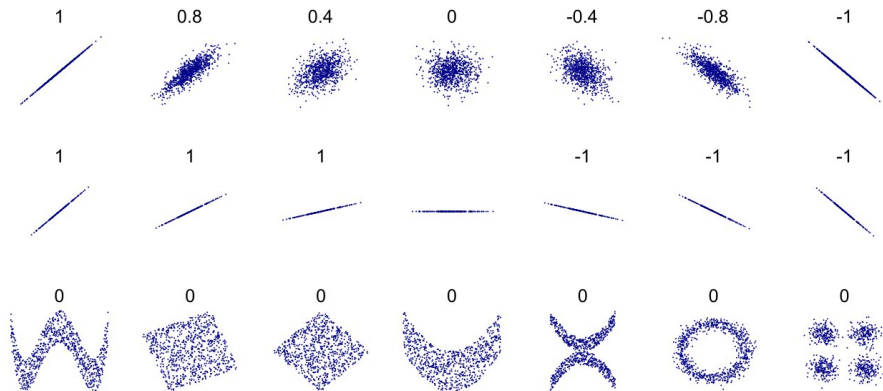


Figure: Visual interpretation of conditional probability

Uncorrelatedness vs. Independence

Independence implies Uncorrelatedness

$$\begin{aligned}\text{Cov}[X, Y] &= E[XY] - E[X] E[Y] = E[X] E[Y] - E[X] E[Y] = 0 \\ \Rightarrow \text{Cor}[X, Y] &= \frac{\text{Cov}[X, Y]}{\sqrt{V[X] V[Y]}} = 0\end{aligned}$$

Uncorrelatedness Does NOT Imply Independence

$$\text{Suppose: } \begin{cases} X \propto U(-1, 1) \\ Y = X^2 \end{cases} \quad \text{Then: } \begin{cases} \text{Cor}[X, Y] = 0 \text{ (Uncorrelated)} \\ X \not\perp Y \end{cases}$$

Correlatedness vs. Causation

Causation Does NOT Imply Correlatedness

Suppose: $\begin{cases} X \propto U(-1, 1) \\ Y = X^2 \end{cases}$ Then: $\begin{cases} \text{Cor}[X, Y] = 0 \text{ (Uncorrelated)} \\ X \text{ clearly causes } Y. \end{cases}$

Correlatedness Does NOT Imply Causation

$\begin{cases} Z \propto U(-1, 1) \\ X = Z^2 \\ Y = Z^2 \end{cases}$ Then: $\begin{cases} \text{Cor}[X, Y] = 1 \text{ (Correlated)} \\ X \text{ and } Y \text{ don't have causal effect on each other.} \end{cases}$

Spurious Correlation [2]



Figure: Violent Crime Index vs Ice Cream Sales

Section 3

Sample Distributions

The Multivariate Gaussian (Normal) Distribution (MVN)

The Multivariate Gaussian (Normal) Distribution

Random vector $\mathbb{Y}$ is said to be multivariate normally distributed if every linear combination of its components has a univariate normal distribution.

Probability Density Function

The PDF for MVN with dimension D is defined as:

$$\mathcal{N}(\mathbf{y}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) \triangleq \frac{1}{(2\pi)^{D/2}|\boldsymbol{\Sigma}|^{1/2}} \exp \left[-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu}) \right]$$

where:

$$\boldsymbol{\mu} = \mathbb{E}[\mathbb{Y}] \in \mathbb{R}^D$$

$$\boldsymbol{\Sigma} = \text{Cov}[\mathbb{Y}] \in \mathbb{R}^{D \times D}$$

MVN Covariance Matrix Properties

Symmetric Matrix

Matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric iff $\mathbf{A} = \mathbf{A}^T$ (We usually show this by $\mathbf{A} \in \mathbb{S}^n$)

Positive (Semi)Definite

Suppose $\mathbf{A} \in \mathbb{S}^n$. Then $\forall \mathbf{v} \in \mathbb{R}^n \setminus \{0\}$:

$\mathbf{A}$ is positive definite (PD), denoted $\mathbf{A} \succ 0$ $\Leftrightarrow \mathbf{v}^T \mathbf{A} \mathbf{v} > 0$

$\mathbf{A}$ is positive semidefinite (PSD), denoted $\mathbf{A} \succeq 0$ $\Leftrightarrow \mathbf{v}^T \mathbf{A} \mathbf{v} \geq 0$

$\mathbf{A}$ is negative definite (ND), denoted $\mathbf{A} \prec 0$ $\Leftrightarrow \mathbf{v}^T \mathbf{A} \mathbf{v} < 0$

$\mathbf{A}$ is negative semidefinite (NSD), denoted $\mathbf{A} \preceq 0$ $\Leftrightarrow \mathbf{v}^T \mathbf{A} \mathbf{v} \leq 0$

$\mathbf{A}$ is indefinite iff it is none of the above.

MVN Covariance Matrix Properties

Covariance Matrix is PSD

Assume Σ to be the covariance matrix of $\mathbb{X}$ D-dimensional random vector. Then:

- $\Sigma \in \mathbb{S}^D$ based on definition.
- $\Sigma \succeq 0$ (PSD) because:

$$\mathbf{v}^T \Sigma \mathbf{v} = V[\mathbf{v}^T \mathbb{X}] \geq 0, \forall \mathbf{v} \in \mathbb{R}^D$$

- If $\mathbb{X}$ is distributed normally, then $\Sigma \succ 0$ (PD) because:

$$\exists \mathbf{v} \neq \mathbf{0} : \mathbf{v}^T \Sigma \mathbf{v} = 0 \rightarrow V[\mathbf{v}^T \mathbb{X}] = 0 \rightarrow \mathbf{v}^T \mathbb{X} \text{ is not normally distributed}$$

Bivariate Normal (D=2)

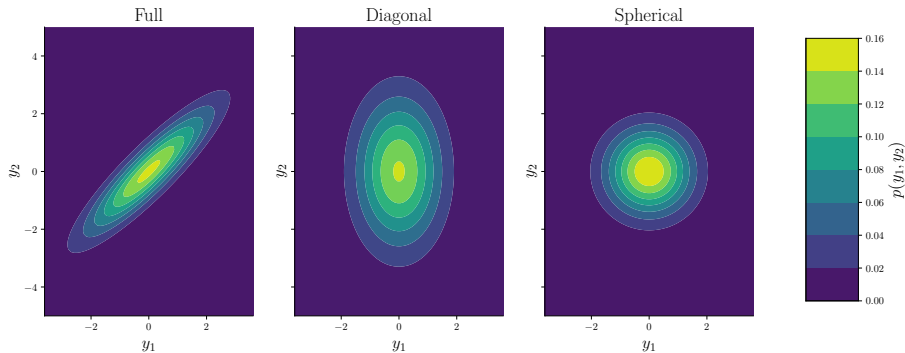


Figure: Level set of constant probability density

Mahalanobis Distance

Mahalanobis Distance

Mahalanobis Distance (Δ) is a metric to calculate the distance between point $\mathbf{y}$ and distribution p with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ and is defined as:

$$\Delta^2 \triangleq (\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu})$$

MVN and Mahalanobis Distance

The log probability of MVN at a specific point $\mathbf{y}$ is given by:

$$\log p(\mathbf{y}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = -\frac{1}{2} \overbrace{(\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu})}^{\Delta^2} + \text{constant}$$

Marginals and Conditionals of an MVN

Suppose $\mathbb{Y} = (\mathbb{Y}_1, \mathbb{Y}_2)$ where $\mathbb{Y}_1$ and $\mathbb{Y}_2$ have D_1 and D_2 dimension, respectively (thus $\mathbb{Y}$ is $(D_1 + D_2)$ -dimensional). Assume $\mathbb{Y}$ to be Gaussian with following parameters:

$$\boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix}, \boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1} = \begin{bmatrix} \boldsymbol{\Lambda}_{11} & \boldsymbol{\Lambda}_{12} \\ \boldsymbol{\Lambda}_{21} & \boldsymbol{\Lambda}_{22} \end{bmatrix}$$

where $\boldsymbol{\mu}_1 \in \mathbb{R}^{D_1}$, $\boldsymbol{\mu}_2 \in \mathbb{R}^{D_2}$, $\boldsymbol{\Sigma}_{ij} \in \mathbb{R}^{D_i \times D_j}$ and $\boldsymbol{\Lambda}_{ij} \in \mathbb{R}^{D_i \times D_j}$. Then the marginals and conditionals are given by:

$$p(\mathbf{y}_1) = \mathcal{N}(\mathbf{y}_1 | \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$$

$$p(\mathbf{y}_2) = \mathcal{N}(\mathbf{y}_2 | \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$$

$$p(\mathbf{y}_1 | \mathbf{y}_2) = \mathcal{N}(\mathbf{y}_1 | \boldsymbol{\mu}_{1|2}, \boldsymbol{\Sigma}_{1|2})$$

where:

$$\boldsymbol{\mu}_{1|2} = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} (\mathbf{y}_2 - \boldsymbol{\mu}_2) \text{ (Affine function of observed vector } \mathbf{y}_2)$$

$$\boldsymbol{\Sigma}_{1|2} = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21} \text{ (Independent of observed vector } \mathbf{y}_2)$$

Using MVN Marginals

Imputing Missing Values

Consider the following scenario:

- Select D movies
- Ask N people to give them scores ($\mathbb{Y} \in \mathbb{R}^D$)
- Some people have not scored all movies.
- You know that the scoring vector comes from $\mathcal{N}(\mathbf{y}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$

How to fill missing scores by MVN marginals?

Solution

We can fill person n scoring vector as:

- Compute $p(\mathbf{y}_{n,\mathbf{h}}|\mathbf{y}_{n,\mathbf{v}}, \boldsymbol{\theta})$ where: $\begin{cases} \boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\Sigma}) : \text{Parameters} \\ \mathbf{h} : \text{missing (hidden) score indices} \\ \mathbf{v} : \text{submitted (visible) score indices} \end{cases}$
- Impute missing values by: $\begin{cases} \bar{\mathbf{y}}_{n,\mathbf{h}} = \mathbb{E}[\mathbb{Y}_{n,\mathbf{h}}|\mathbf{y}_{n,\mathbf{v}}, \boldsymbol{\theta}] : \text{Posterior mean} \\ \text{Posterior sampling} \end{cases}$

Imputing Missing Values

How to estimate $\boldsymbol{\mu}$ and $\boldsymbol{\theta}$? *Solution:* By using *Expectation Maximization*.

Section 4

Linear Gaussian Systems

Linear Gaussian Systems (LGS)

Linear Gaussian Systems

Assume the following items:

- $\mathbf{Z} \in \mathbb{R}^L$: Unknown vector
- $\mathbf{Y} \in \mathbb{R}^D$: Noisy measurements
- The following distributions hold:
 - $p(\mathbf{z}) = \mathcal{N}(\mathbf{z} | \boldsymbol{\mu}_z, \boldsymbol{\Sigma}_z)$
 - $p(\mathbf{y} | \mathbf{z}) = \mathcal{N}(\mathbf{y} | \mathbf{W}\mathbf{z} + \mathbf{b}, \boldsymbol{\Sigma}_y)$, $\mathbf{W} \in \mathbb{R}^{D \times L}$, $\mathbf{b} \in \mathbb{R}^D$

then:

- Joint distribution $p(\mathbf{z}, \mathbf{y}) = p(\mathbf{z})p(\mathbf{y} | \mathbf{z})$ is a $L + D$ dimensional Gaussian with the following parameters:

$$\boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_z \\ \mathbf{W}\boldsymbol{\mu}_z + \mathbf{b} \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_z & \boldsymbol{\Sigma}_z \mathbf{W}^T \\ \mathbf{W}\boldsymbol{\Sigma}_z & \boldsymbol{\Sigma}_y + \mathbf{W}\boldsymbol{\Sigma}_z \mathbf{W}^T \end{bmatrix},$$

- Using Bayes rule, the posterior $p(\mathbf{z} | \mathbf{y})$ is also L dimensional Gaussian with the following parameters:

$$\begin{aligned} \boldsymbol{\Sigma}_{z|y}^{-1} &= \boldsymbol{\Sigma}_z^{-1} + \mathbf{W}^T \boldsymbol{\Sigma}_y^{-1} \mathbf{W} \\ \boldsymbol{\mu}_{z|y} &= \boldsymbol{\Sigma}_{z|y} \left[\mathbf{W}^T \boldsymbol{\Sigma}_y^{-1} (\mathbf{y} - \mathbf{b}) + \boldsymbol{\Sigma}_z^{-1} \boldsymbol{\mu}_z \right] \end{aligned}$$

Conjugate Priors

Conjugate Priors

Assume $\mathcal{F}$ as a family of distribution functions (*e.g.* Gaussian). We say that a prior $p(\mathbf{z}) \in \mathcal{F}$ is a conjugate prior for a likelihood function $p(\mathbf{y}|\mathbf{z})$ if the posterior is in the same family of distribution, i.e., $p(\mathbf{z}|\mathbf{y}) \in \mathcal{F}$.

Conjugate Priors

Based on slide 22, Gaussian prior is a conjugate prior for the Gaussian likelihood.

Linear Gaussian System

Inferring an Unknown Scalar

Suppose:

- *Prior*: We want to estimate unknown quantity Z where $p(z) = \mathcal{N}(z|\mu_0, \lambda_0^{-1})$
- *Likelihood* We have N independent noisy measurements y_i distributed as $p(y_i|z) = \mathcal{N}(y_i|z, \lambda_y^{-1})$

compute the posterior $p(z|y_1, \dots, y_N)$.

Solution

We start by defining $\mathbb{Y} = (y_1, \dots, y_N)$. Then we can easily show that the problem is linear Gaussian system with $\mathbf{W} = \mathbf{1}_N$ and $\Sigma_y^{-1} = \text{diag}(\lambda_y \mathbf{I})$. Thus:

$$p(z|\mathbf{y}) = \mathcal{N}(z|\mu_N, \lambda_N^{-1})$$

where:

$$\begin{aligned}\Sigma_{z|\mathbf{y}}^{-1} &= \Sigma_z^{-1} + \mathbf{W}^T \Sigma_y^{-1} \mathbf{W} \Rightarrow \lambda_{z|\mathbf{y}} = \lambda_0 + \mathbf{1}^T \text{diag}(\lambda_y \mathbf{I}) \mathbf{1} = \lambda_0 + N\lambda_y \\ \mu_{z|\mathbf{y}} &= \Sigma_{z|\mathbf{y}} \left[\mathbf{W}^T \Sigma_y^{-1} (\mathbf{y} - \mathbf{b}) + \Sigma_z^{-1} \mu_z \right] \Rightarrow \mu_{z|\mathbf{y}} = \lambda_{z|\mathbf{y}}^{-1} \left[\mathbf{1}^T \text{diag}(\lambda_y \mathbf{I}) (\mathbf{y} - \mathbf{0}) + \lambda_0 \mu_0 \right] \\ \Rightarrow \mu_{z|\mathbf{y}} &= \frac{N\lambda_y \bar{y} + \lambda_0 \mu_0}{\lambda_{z|\mathbf{y}}} = \frac{N\lambda_y}{N\lambda_y + \lambda_0} \bar{y} + \frac{\lambda_0}{N\lambda_y + \lambda_0} \mu_0\end{aligned}$$

Linear Gaussian System

LGS system with $N = 1, \lambda_y = 1.0$

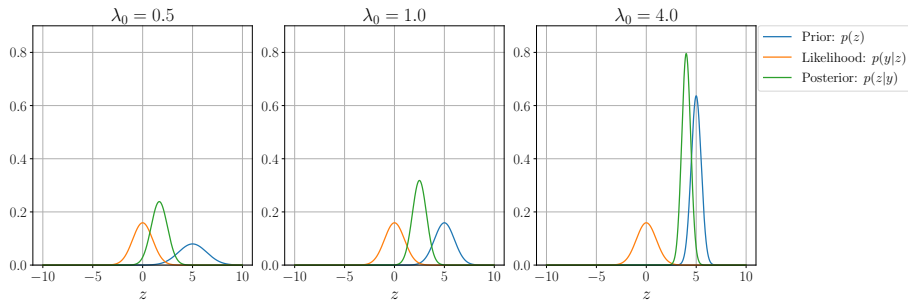


Figure: Prior precision (λ_0) effect

Linear Gaussian System

LGS system with $N = 1, \lambda_0 = 1.0$

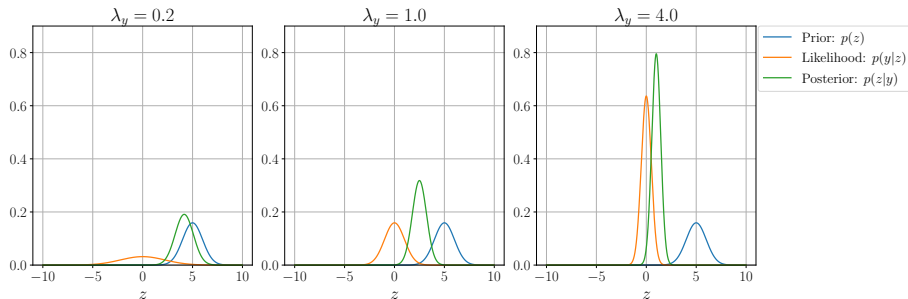


Figure: Likelihood precision (λ_y) effect

Linear Gaussian System

LGS system with $\lambda_0 = 1.0$, $\lambda_y = 1.0$

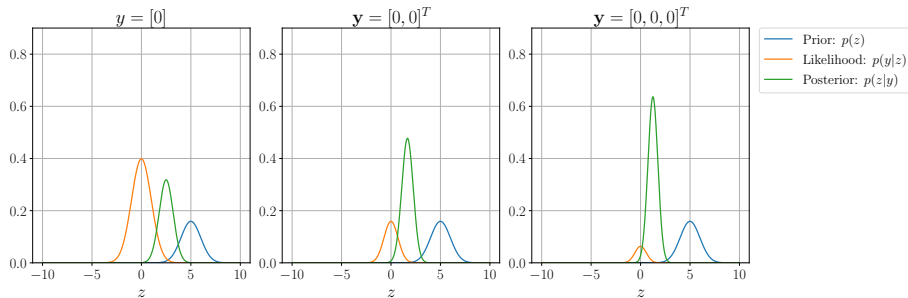


Figure: Number of measurements (N) effect

Sensor Fusion

Suppose:

- *Prior*: We want to estimate unknown vector $\mathbb{Z}$ where $p(\mathbf{z}) = \mathcal{N}(\mathbf{z}|\mu_0, \Sigma_0)$
- *Likelihood*: We have 2 sensors and 1 measurements of each sensor, denoted $\mathbb{Y}_1$ and $\mathbb{Y}_2$, distributes as $\mathcal{N}(\mathbf{y}_i|\mathbf{z}, \Sigma_i)$ (Σ_i demonstrates the reliability for i -th sensor).

compute the posterior $p(\mathbf{z}|\mathbf{y}_1, \mathbf{y}_2)$.

Solution

We start by defining $\mathbb{Y} = (\mathbb{Y}_1, \mathbb{Y}_2)$. Then we can easily show that the problem is linear Gaussian system with $\mathbf{W} = [\mathbf{I}; \mathbf{I}]$ and $\Sigma_y = \begin{bmatrix} \Sigma_1 & \mathbf{0} \\ \mathbf{0} & \Sigma_2 \end{bmatrix}$. Thus the posterior $p(\mathbf{z}|\mathbf{y}) = \mathcal{N}(\mathbf{z}|\mu_{z|y}, \Sigma_{z|y})$ where $\mu_{z|y}$ and $\Sigma_{z|y}$ can be calculated using formulas in Slide 22.

Sensor Fusion

Suppose the sensor fusion example in Slide 28, with the following parameters:

$$\mu_0 = [0; 0], \quad \Sigma_0 = 1000\mathbf{I}, \quad \Sigma_1 = \Sigma_2 = 0.01\mathbf{I}$$

and assume $\mathbf{y}_1 = (0, -1)$ and $\mathbf{y}_2 = (1, 0)$. Visualize the measurements and posterior $p(\mathbf{z}|\mathbf{y})$.

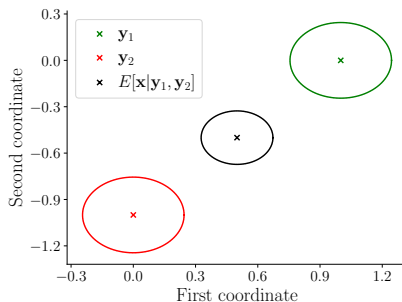


Figure: Sensor fusion result

Sensor Fusion

Suppose the sensor fusion example in Slide 28, with the following parameters:

$$\boldsymbol{\mu}_0 = [0; 0], \quad \boldsymbol{\Sigma}_0 = 1000\mathbf{I}, \quad \boldsymbol{\Sigma}_1 = 0.01\mathbf{I}, \quad \boldsymbol{\Sigma}_2 = 0.05\mathbf{I}$$

and assume $\mathbf{y}_1 = (0, -1)$ and $\mathbf{y}_2 = (1, 0)$. Visualize the measurements and posterior $p(\mathbf{z}|\mathbf{y})$.

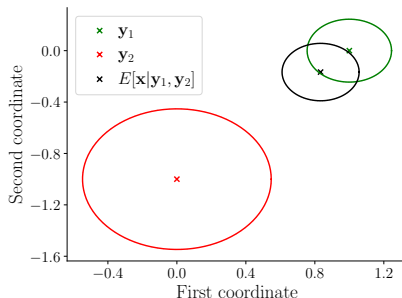


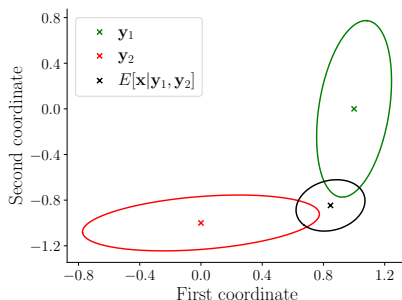
Figure: Sensor fusion result

Sensor Fusion

Suppose the sensor fusion example in Slide 28, with the following parameters:

$$\boldsymbol{\mu}_0 = [0; 0], \quad \boldsymbol{\Sigma}_0 = 1000\mathbf{I}, \quad \boldsymbol{\Sigma}_1 = 0.01 \begin{bmatrix} 10 & 1 \\ 1 & 1 \end{bmatrix}, \quad \boldsymbol{\Sigma}_2 = 0.01 \begin{bmatrix} 1 & 1 \\ 1 & 10 \end{bmatrix}$$

and assume $\mathbf{y}_1 = (0, -1)$ and $\mathbf{y}_2 = (1, 0)$. Visualize the measurements and posterior $p(\mathbf{z}|\mathbf{y})$.



Section 5

Mixture Models

Mixture Models

Mixture Models

One way to create more complex probability models is to take a convex combination of simple distributions. This is called a mixture model. This has the form $p(\mathbf{y}|\boldsymbol{\theta}) = \sum_{k=1}^K \pi_k p_k(\mathbf{y})$ where:

- p_k is the k -th mixture component
- $\{\pi_k\}_{k=1}^K$ are mixture weights with the following constraints:
 - $0 \leq \pi_k \leq 1, k = 1, \dots, K$
 - $\sum_{k=1}^K \pi_k = 1$

Mixture Models - Generative Story

Suppose latent variable Z to be a categorical RV and distributed as $p(z|\boldsymbol{\theta}) = \text{Cat}(z|\boldsymbol{\pi})$ and conditional $p(\mathbf{y}|z = k, \boldsymbol{\theta}) = p_k(\mathbf{y}) = p(\mathbf{y}|\boldsymbol{\theta}_k)$. We can interpret mixture models as follows:

- We sample a specific component.
- We generate $\mathbf{y}$ using sampled value of z .

Using the above procedure, we have:

$$p(\mathbf{y}|\boldsymbol{\theta}) = \sum_{k=1}^K p(z = k|\boldsymbol{\theta})p(\mathbf{y}|z = k, \boldsymbol{\theta}) = \sum_{k=1}^K \pi_k p(\mathbf{y}|\boldsymbol{\theta}_k)$$

Gaussian Mixture Model

Gaussian Mixture Model

Gaussian Mixture Model (GMM) or Mixture of Gaussian (MoG) is defined as:

$$p(\mathbf{y}|\boldsymbol{\theta}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{y}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

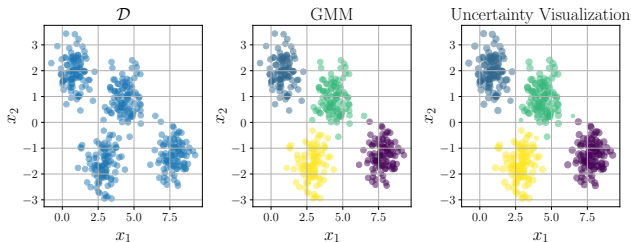


Figure: Sample GMM distribution and its application for clustering

References I



“Pearson correlation coefficient,”

https://en.wikipedia.org/wiki/Pearson_correlation_coefficient.



“The logic of causal conclusions: How we know that fire burns, fertilizer helps plants grow, and vaccines prevent disease,”

<http://icbseverywhere.com/blog/2014/10/the-logic-of-causal-conclusions/>.